Problem Set 5

Fundamentals of Numerical Programming I
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January 12, 2026

Problem 1. Adaptive Quadrature

Consider the following functions

$$f_1(x) = \exp\left(\frac{1}{2}x\right), \quad f_2 = \sqrt{1+x^4}, \quad f_3 = \ln\left(1+\exp(x)\right) \tag{1}$$

We are interested in integrals $(a, b \in \mathbb{R}, a < b)$

$$\int_a^b f_i(x)\,dx \tag{2}$$

f_1 is trivial to integrate, f_2 and f_3 have no elementary integrals.

In the lecture we have discussed the **midpoint rule**

$$M_{[a,b]}(f) := (b-a)f\left(\frac{a+b}{2}\right) \tag{3}$$

the **trapezoidal rule**

$$T_{[a,b]}(f) := \frac{b-a}{2}\left(f(a)+f(b)\right) \tag{4}$$

and **Simpson's rule**

$$S_{[a,b]}(f) := \frac{b-a}{6}\left(f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right) \tag{5}$$

(here written for the whole interval rather than for sub-intervals as in the lecture).

One can show that for a convex function f $(f''(x) \geq 0)$

$$M_{[a,b]}(f) \leq \int_a^b f_i(x)\,dx \leq T_{[a,b]}(f) \tag{6}$$

The second inequality follows from f being under the tangent for a convex f and the first is a form of *Jensen's inequality.*

This allows us to construct an **adaptive quadrature scheme**

$$Q_{[a,b],\epsilon}(f) := \begin{cases} S_{[a,b]}(f), & \text{if } |M_{[a,b]}(f) - T_{[a,b]}(f)| < \epsilon \\ Q_{\left[a,\frac{a+b}{2}\right],\frac{\epsilon}{2}}(f) + Q_{\left[\frac{a+b}{2},b\right],\frac{\epsilon}{2}}(f), & \text{else} \end{cases} \tag{7}$$

with a given error bound $\epsilon \in \mathbb{R}_{>0}$, such that

$$\left| \int_a^b f(x)\,dx - Q_{[a,b],\epsilon}(f) \right| < \epsilon \tag{8}$$

(a) Show that f_1, f_2 and f_3 are globally convex, i.e. $\forall x \in \mathbb{R} : f_i''(x) \geq 0, i = 1, 2, 3.$

(b) Implement the adaptive quadrature scheme outlined above. Calculate

$$\int_0^1 f_i\,dx, \quad i = 1, 2, 3 \tag{9}$$

with a precision of $\epsilon = 10^{-4}$. Validate your result for f_1 against the analytical solution.

Should you finish very quickly, you are invited to work on the optional exercises from the previous days :)